

Engage Phase – Axis 5: Brain Load & Hidden Effort

Summary of Key Deliverables

Within the challenge “Make the invisible visible in brain imaging,” Axis 5 focuses on detecting hidden cognitive effort using resting-state fMRI and functional connectivity.

The core idea is that a brain may appear structurally normal on MRI while still being functionally imbalanced. This can result in symptoms such as fatigue or cognitive overload that are not explained by conventional imaging.

Essential Question:

What if patients with normal MRI scans are actually experiencing hidden brain overload?

Challenge:

Reveal hidden cognitive effort using fMRI-based connectivity and interpretable AI models.

This axis addresses a key clinical gap: many patients report symptoms without visible abnormalities, making diagnosis and explanation difficult. By shifting from structure to function, we focus on how brain regions interact rather than how they appear.

We approach the brain as a network, where dysfunction can manifest through altered connectivity patterns such as hyperconnectivity or reduced efficiency.

Target populations include patients with cognitive fatigue, post-stroke compensation, or unexplained neurological symptoms. Stakeholders include clinicians, patients, and AI researchers.

Outcome:

The Engage phase allowed us to clearly define the problem, establish a focused challenge, and set a direction centered on functional connectivity and clinically relevant AI tools.